



Enhanced cutaneous wound healing by *Senecio serratuloides* (Asteraceae/Compositae) in a pig model

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ABSTRACT

Senecio serratuloides (Asteraceae/Compositae) is a traditional herbal remedy used to treat skin wounds in South Africa. The plant contains hepatotoxic alkaloids and is responsible for the death of farm animals which have eaten the plant. Wound healing efficacy was determined by applying prepared plant material to deep partial-thickness wounds on a pig model and healing was compared to controls over 16 days. Regrowth of the skin was monitored by measuring the increasing ratio of the epidermal thickness, relative to that of normal skin. The neo-epidermis of plant-treated wounds was significantly thicker on days 5 and 7 and was maximal on day 7, compared with day 9 for the controls. The pH of the treated wound surface was significantly lower than the controls on day 2. The presence of alkaloids was confirmed with Dragendorff's reagent on thin layer chromatography, but liver function enzymes were no different to that of the controls and no signs of liver toxicity were apparent on histology when the plant was applied topically to wounds. *S. serratuloides* treatment accelerated the healing of deep partial thickness wounds and the epidermal thickness ratio was found to be a useful marker in monitoring the wound healing process. The significance of the lower wound surface pH remains to be fully determined.

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1. Introduction

Treating and caring for wounds place an enormous, a significant and escalating burden on hospital and health care resources, with an estimated cost in 2010 of \$15.3 billion in the USA alone (Sen et al., 2009). Sophisticated care is often unaffordable in developing countries, where many patients rely on familiar, readily accessible and inexpensive traditional wound healing remedies. In South Africa, the herb, *Senecio serratuloides* (Asteraceae/Compositae), the “2 day cure”, is reported to promote the healing of cuts, sores, and burns (van Wyk et al., 1997). Fresh, dried or even charred leaves are applied directly onto the wound, which may then be covered with pig or zebra fat. Fawole et al. (2010) have demonstrated anti-oxidant and other beneficial properties of the plant. Despite the widespread use of this plant, its wound healing efficacy and mechanisms have yet to be subjected to rigorous scientific investigation.

Moreover, consumption of this plant in various situations has led to hepatic, renal and pulmonary disorders. This toxicity has been attributed to the presence of certain pyrrolizidine alkaloids (PA) (McLean, 1970) found within plants of the genus *Senecio*, although these have not specifically been reported in *S. serratuloides*. We are unaware of

any reported cases of PA poisoning following the topical application of *S. serratuloides* to skin wounds. AMES and VITOTOX® tests of solvent extracts of this plant have shown no signs of mutagenicity, except at concentrations exceeding 2000 µg/ml (Elgorashi et al., 2003).

Therefore the aims of this study were to a) identify the presence of alkaloids in the plant; b) to establish efficacy of the *S. serratuloides* in healing deep partial thickness cutaneous wounds in a pig model; and c) to determine whether the topical application of *S. serratuloides* produces any signs of toxicity in the test animals. In addition, we measured wound pH which has been proven to significantly influence the process and phases of wound healing (Schneider et al., 2007). Furthermore, we propose the measurement of the ratio of the wound-to-normal skin epidermal thickness as a useful parameter in assessing the rate of wound re-epithelialisation.

2. Methods

2.1. Collection and preparation of plant material

The plant material, purchased between 2008 and 2010 from the Mooi Street Traditional Market, Johannesburg, was identified and deposited at the C.E. Moss Herbarium of the School of Animal, Plant and Environmental Sciences, University of the Witwatersrand. Voucher numbers were 08/05/2009, 09/08/2009, 09/05/2009 and 27/02/2010.

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The material was air dried and stored frozen at -20°C until used. The preparation and application of the plant material were according to the traditional healer's instructions. Leaves were dried in an oven at 40°C overnight, after which they were crushed to a fine powder using a pestle and mortar and then heated on a hot plate until charred to a black charcoal-like "ash". The material was ground again and this prepared plant material was screened for alkaloids and placed directly on the wounds (described below).

2.2. Phytochemical screening for alkaloids

The presence of alkaloids in the plant was determined (Harborne, 1988) by extracting 20 g of prepared plant material (dried, ground and charred) with 200 ml of water/methanol (Merck, South Africa; vol/vol) for 2 days. After sonicating the mixture for 1 h, the extract was filtered under vacuum to remove the residual plant material and then evaporated to 10% of the original volume at 60°C over 24 h. The pH of the concentrated extract was adjusted to pH 3 with 1 M hydrochloric acid and washed with chloroform to remove polar compounds. The chloroform layer was discarded and the pH re-adjusted to 8.0 with 1 M sodium hydroxide. The alkaloid components were then extracted into dichloromethane and separated on thin layer chromatography (TLC) plates (Sigma, South Africa) with hexane and ethyl acetate (Merck, South Africa) (vol:vol) as the mobile phase. The TLC plates were visualized using ultraviolet light, iodine vapour and stained with cerium sulphate and Dragendorff's reagent.

2.3. Animal model and test groups

Large white pigs (*Sus scrofa domesticus*) were selected for experimentation because their skin is similar to human skin (Sullivan et al., 2001). Ethical clearance was obtained from the Animal Ethics Committee (Clearance number: 2008/15/04) of the University of the Witwatersrand.

2.4. Surgery and wound generation

Nine ($n = 9$) female pigs (± 30 kg) were anaesthetized using injectable midazolam (0.3 mg/kg) (Dormicum, Roche) and ketamine (11 mg/kg) (AnaketV; Bayer HealthCare). Anaesthesia was maintained by Isofor (Bayer HealthCare) as the volatile agent. Dr Patel performed the surgery at the Central Animal Unit in the Faculty of Health Sciences. Deep partial thickness wounds were created with an electric dermatome, set to 800 μm (Aesculap HH Dermatome Ga 630) to remove 6 square sections of skin (2.5 cm $\times$ 2.5 cm) spaced

4 cm apart and either side of the midline on the back of each pig (Fig. 1).

2.5. Treatment and dressing

The wounds were treated with one of the following: 1) prepared *S. serratuloides* material (0.2 g); 2) activated carbon (0.2 g, Merck South Africa; Cat. No. 102186) as a positive control; or 3) left untreated as the negative control. Activated carbon was used as a positive control as a commercially available wound dressing product is available (Smith and Nephew's Actisorb). We used a Merck activated charcoal product of similar particle size and consistency to the charcoal-like powder prepared from the plant. The weight of charcoal used was equal to the prepared plant material used on each of the wounds. Each treated wound was also separately covered with an occlusive dressing integrated with an absorptive pad (3 M Tegaderm® Cat. No. 3584). The test area of each pig was wrapped in a stocking to hold the dressings in place. Each pig was housed separately in the Central Animal Unit and buprenorphine (0.3–0.6 ml) (Temgesic, Reckitt Benckiser Pharmaceuticals) was administered as post-operative analgesia.

2.6. Sample collection and analysis

The dressings were changed on days 2, 5, 7, 9, 12, and 16 based on the schedule of the Central Animal Unit. At each dressing change, wound surface pH was measured, and photographs, skin biopsies and jugular venous blood were taken. Liver biopsies were collected 1 month post-completion of the experiments.

2.7. Skin biopsy collection and analysis

Biopsies taken for histological analysis were immediately placed in 10% buffered formalin for 48 h, dehydrated in a graded series of ethanol and finally embedded in paraffin wax. Sections were cut at 5–7 μm on a sledge microtome. The sections were stained with Haematoxylin and Eosin (H and E) using standard methods, photographed at 100 $\times$ magnification and analysed with planimetry software (IMAGEJ). At each time point, on each section, 10 measurements of the thickness of the epidermis were taken a) inside the recovering wound space and b) from normal, uninjured reference tissue, adjacent to the wound area. An "epidermal thickness ratio" was calculated by dividing the epidermal thickness measurement within the wound by that of the uninjured skin. A ratio close to 1 would indicate that the injured skin's epidermal thickness was similar to that of normal skin.

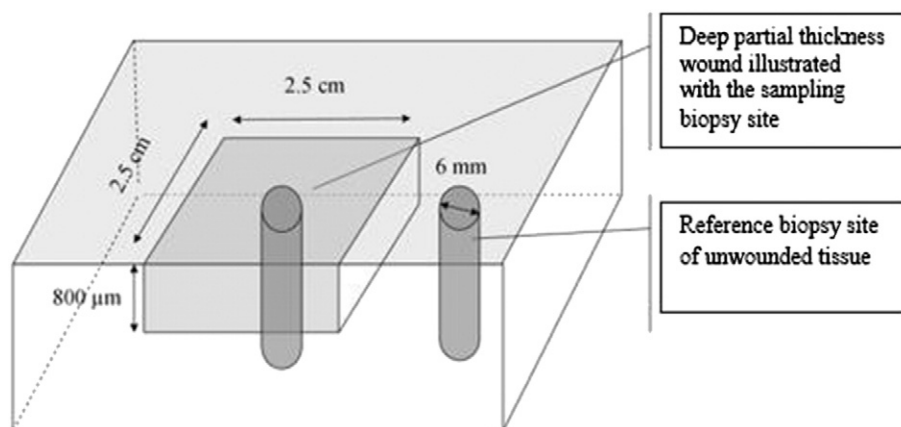


Fig. 1. A schematic diagram of one of the 2.5 cm by 2.5 cm deep partial thickness experimental wounds, 800 μm in depth, created on the back of the pig. The location of the core biopsy sites, relative to the wound area, is illustrated. A biopsy of normal uninjured skin, outside the wound area, was taken as the control external reference tissue.

2.8. pH measurement

On the days of sample collection, the wound pH was measured using a surface electrode (Separations Scientific ST P17-BNC) and the average of 3 measurements were plotted against the post-operative time in days.

2.9. Analysis of liver hepatotoxicity

Blood was collected from the jugular vein on the specified days for liver function tests (LFTs) in 8.5 ml yellow top SST gel tubes (BD Biosciences, South Africa (Cat. No. 367955) and sent to Contract Lab Services (CLS, Wits Health Consortium, Johannesburg) for analysis. One month after the completion of the experiments, the animals were euthanased and liver biopsies were taken from 2 lobes of the liver. The samples were placed in 10% buffered formalin and sent to Golden Vet Pathologists at IDEXX Laboratories (Onderstepoort, Pretoria, South Africa) to screen for symptoms of PA poisoning.

2.10. Statistical analysis

Results were reported as mean $\pm$ standard error of mean (SEM). Statistical significance was determined using Statistica (Version 9). Analysis between the treatment groups was determined using the Kruskal–Wallis test, with a *post-hoc* analysis.

3. Results

3.1. Phytochemical screening

Ultra-violet light and iodine visualized spots on the TLC plates with Rf values ranging from 0.2 to 0.72. However, Dragendorff's reagent visualized a single spot (Rf = 0.72), whereas cerium sulphate failed to produce any positive results.

3.2. Subjective gross and histological analysis

At day 2, the wounds appeared similar with fresh wound beds easily visualized after the residual plant and activated carbon material were removed (images not shown) and no epidermis could be seen on

histology in any of the treatment groups. However, by day 5, newly formed epidermis could be seen in all three treatment groups (Fig. 2). By day 7 post-operatively, the epidermis in the *S. serratuloides* group appeared thicker than that of the controls (Fig. 3) and was more resilient to debriding.

In all treatment groups the epidermal ratio increased, exceeding a value of 1, and then declined until day 16 (Fig. 4). On day 5, there was a significant difference between the activated carbon treatment and the *S. serratuloides* ($p = 0.00018$), as well as the negative control wounds ($p = 0.0046$), but without differences between the two latter treatments ($p = 0.33$). By day 7, the epidermal ratio in the *S. serratuloides* group had peaked 2 days earlier than the activated carbon and negative control groups. The epidermis in the *S. serratuloides* treatment group was significantly thicker than that of the activated carbon ($p = 0.0036$) and negative control ($p = 0.0052$) groups. Once all treatments had peaked in thickness, they became thinner and from day 9 onward, there were no significant differences between any of the treatment groups.

3.3. pH measurements

The pH (Fig. 5) of fresh wounds was initially between 7.93 and 7.97. After treating the wounds the pH increased to 8.02 for the experimental plant, 8.37 for the activated carbon and 8.41 for the negative control groups 2 days post-operatively. The pH of the *S. serratuloides* group was significantly lower compared to the activated carbon ($p = 0.001$) and negative control groups ($p = 0.013$). Thereafter, the pH decreased at similar rates in all groups. The rate of pH decline with time changed around day 5 postoperatively, with all treatments ending with a similar pH by post-operative day 16.

3.4. Liver function tests

For all treatment groups, LFTs (data not represented) show that all measured liver enzymes were raised beyond the normal ranges as described by Merck Veterinary Guidelines. Biopsy specimens collected from the pigs' livers were negative for any sign of acute or chronic PA poisoning or toxicity.

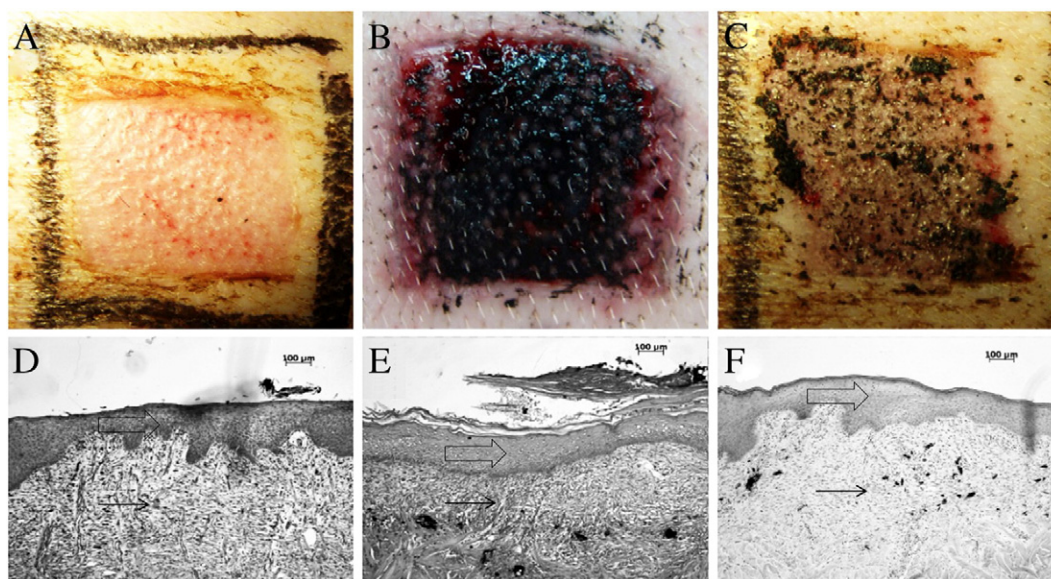


Fig. 2. Wounds viewed on day 5 post-operative with the negative control (A), activated carbon (B), and *Senecio* (C). Corresponding histological sections are seen in D, E, and F being negative control (D), activated carbon (E), and *Senecio* (F). In the gross images (A, B and C) a prominent epithelial layer is present in the *Senecio* treated wounds, which was not evident in the control. In the histological sections, the arrows that are not filled indicate the regenerating epidermis, which was not evident in the corresponding gross images of the control wounds. The solid arrows indicate the granulation tissue deep to the epidermis. Histological sections were imaged at 100 $\times$ magnification.

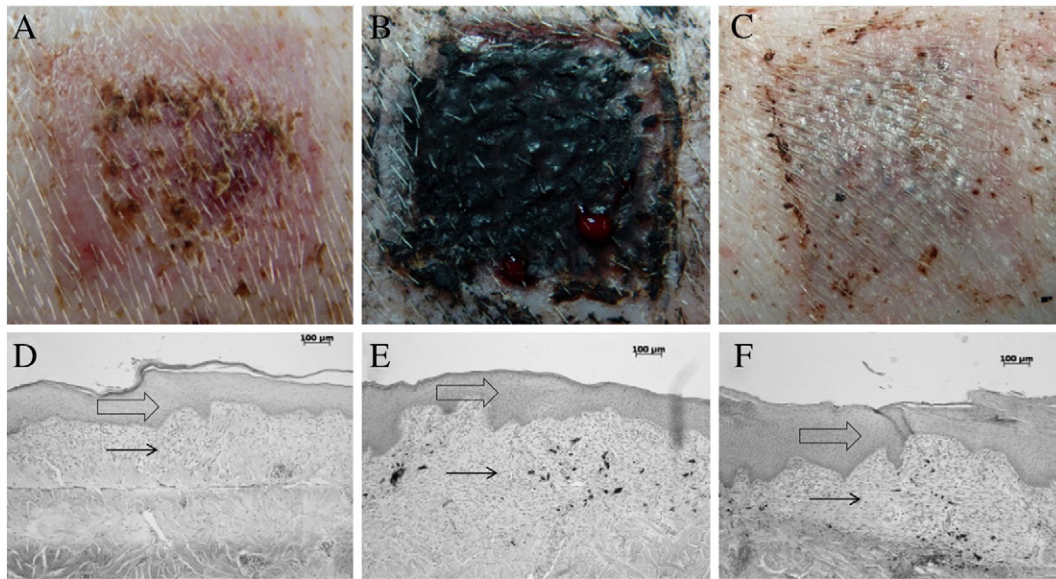


Fig. 3. Wounds viewed on day 7 post-operative with the negative control (A), activated carbon (B), and *Senecio* (C). Corresponding histological sections are seen in D, E, and F being negative control (D), activated carbon (E), and *Senecio* (F). In the gross images (A, B and C) a prominent eschar was seen in the control wounds (A and B) but was not seen in the *Senecio* treated wounds. In the histological sections, the arrows that are not filled indicate the regenerating epidermis, which was thicker in the *Senecio* treated wounds. The solid arrows indicate the granulation tissue deep to the epidermis. Histological sections were imaged at 100× magnification.

4. Discussion

The overall finding of this study was that *S. serratuloides*, prepared in a traditional manner, healed deep partial thickness skin wounds in a pig model within 5 days, compared to 7 days with wounds treated with either activated carbon or dressing alone. This study, measuring qualitative gross and histological parameters of wound recovery over 16 days, complements an earlier study on this plant, which tested extracts for genotoxicity, anti-inflammatory, anti-cholinesterase, and anti-oxidant properties (Fawole et al., 2010).

The increase, and then subsequent decrease, of the epidermal ratio may follow the sequential phases of wound healing proliferation and maturation. The maximum measurement of the epidermal ratio for *S. serratuloides* treated wounds was 2 days earlier than that of the other groups. All 3 treatment groups, however, had a similar epidermal thickness ratio by day 16. The plant therefore appears to affect the early phases of wound healing, causing either accelerated proliferation, earlier maturation of the epidermal layers, or a combination of these.

Previous studies have only measured the actual epidermal thickness, at a single time point (Lynch et al., 1987). The epidermal thickness ratio, while being novel, also provides a measure of the completeness of the wound healing process, as compared to uninjured skin. Furthermore, the ratio accounts for the variable thickness of the epidermis of normal uninjured skin. During the study, however, the epidermal thickness ratio did not reach a “normal” ratio of 1, which suggested that the observational study period was not long enough for wound maturation and remodelling to be completed. Measures such as an epidermal thickness ratio, while providing a gross measure of healing, do not measure the functional state of the epidermis and other additional measures are needed to fully evaluate the efficacy of wound healing treatments.

The overall integrity of intact skin requires appropriate pH to maintain homeostasis and the correct balance of proteases, which are important for the removal of cellular debris and the modulation of growth factors, cytokines and cytoplasmic receptor domains (Yabkowitz, et al., 1999 and Ossovskaya and Bunnet, 2004). Additionally, a lower pH has been shown to promote cellular migration and proliferation (Schneider

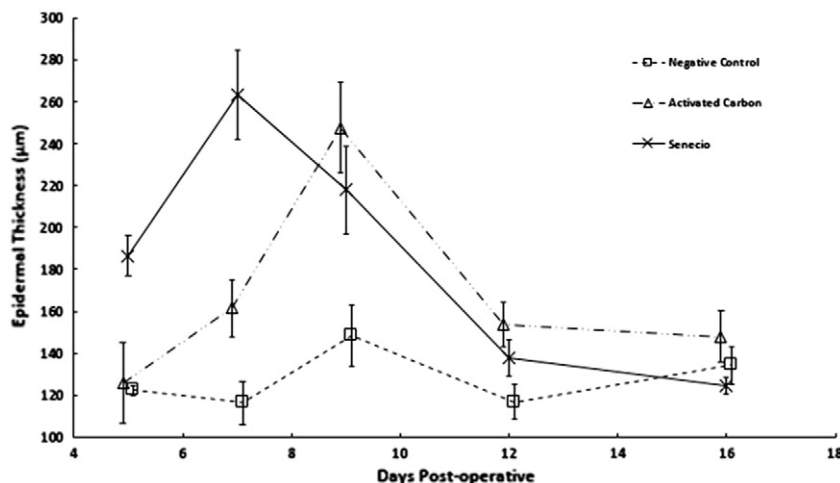


Fig. 4. Epidermal thickness ratio (mean) of the deep partial thickness wounds plotted against time (days post-operative). Error bars represent the SEM.

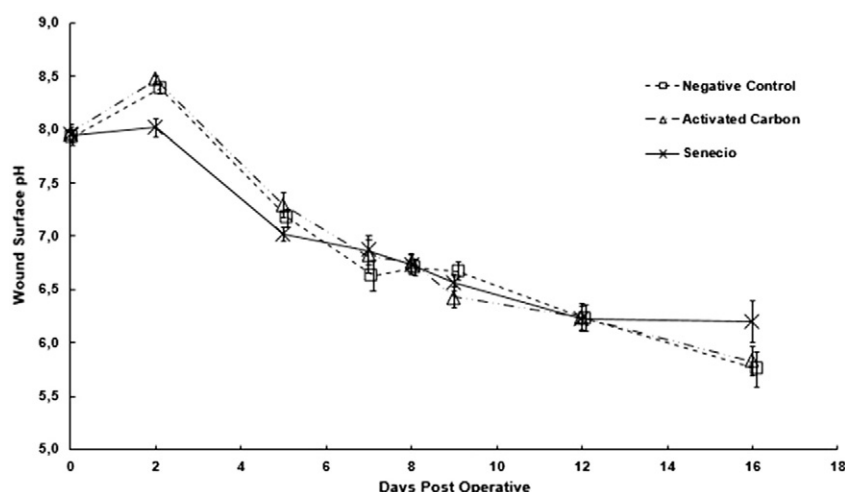


Fig. 5. Wound pH plotted against time (days post-operative). Error bars represent the SEM.

et al., 2007). This simple measure was determined to follow the wound healing process of deep partial thickness wounds in this investigation.

The pH of the *S. serratuloides* treated wounds was initially significantly lower (more acidic) than the controls, after which the decrease in pH was similar, without differences between the treatment groups. It is possible that the lower pH at day 2 may have had an effect on the biochemistry of the healing wounds which will require further investigation. Furthermore, from day 12 the pH of the *S. serratuloides* treated wounds normalized, suggesting a return to normal function. Further analysis is required to determine the significance of these observations.

The South African National Biodiversity Institute (SANBI, 2010) has classified *S. serratuloides* as a poisonous plant because hepatotoxicity has been noted subsequent to accidental ingestion (McLean, 1970). In this study, topical application of the plant material did not produce histological signs of hepatotoxicity. PAs may exist naturally as either the *N*-oxide form or a more toxic free base form and when the plants are ingested, the *N*-oxides could be converted to the free base form by the acid within the gastrointestinal tract (Mattocks, 1971). In the present study, the PAs would not readily convert to the free base when administered percutaneously (Brauchli, et al., 1982) possibly due to them bypassing the acidic environment of the stomach and this may explain the lack of toxicity reported in this study. Further work will be required to determine whether this hypothesis is correct. Additionally, not all PAs are toxic (Rösemann et al., 2014) or present in the every *Senecio* plant and this would require further investigation using chromatographic techniques.

Despite the evidence suggesting toxicity following oral administration, Toma et al. (2004) successfully treated ethanol-induced gastric ulcers with a PA-containing extract of *S. brasiliensis*. This study showed that epidermal growth factor (EGF) was elevated in the treated tissues, indicating a possible mechanism to explain the healing properties of this genus of plants (Toma et al., 2004). Interestingly, anecdotal reports have emerged from the traditional healing community that *S. serratuloides* has been used to treat gastric ulcers. Furthermore, as a percentage of total skin surface area, the extent of the experimental wounds created was small compared to that which could be encountered in a clinical setting, for example, a burn wound or skin graft and therefore toxic symptoms may be sub-clinical.

In the present study, it must be noted that neither the chemical composition nor concentration of the active components or PAs, applied to the wounds was known. Although the phytochemical screen identified the presence of alkaloids in the plant, Dragendorff's reagent is not specific for PAs and additional analysis would be needed to confirm their presence.

The use of this traditional medicine as a charred product may not be acceptable to certain patients and the concentration of the active components may be subject to seasonal variation. Therefore, additional variables need to be measured to determine the mechanism and concentrations at which the active components promote proliferation or maturation of the epidermal layers. This would require the isolation of the active compounds, which would a) facilitate identifying the biochemical pathways involved; b) allow concentrations of the active components to be optimized to reproducibly promote wound healing; c) allow appropriate toxicity studies to be conducted and d) the development of potential delivery systems. Furthermore, the ability of this plant or its isolated active components to treat other wound models needs to be determined.

5. Conclusion

S. serratuloides treatment was found to be effective in enhancing the wound healing process of deep partial thickness wounds in a pig model. The epidermal thickness ratio and wound pH were included in our study as additional markers of the wound healing process and environment. Under the experimental conditions used, histological signs of liver toxicity were not apparent. Further studies are needed to elucidate the mechanism by which this plant promotes wound healing.

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